Daniel Boyland

Biomedical Scientist · Clinical Education & Quality Specialist

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Core specialisms

ISO, quality & technical

- ISO 15189, 17025 & 17043 (train, assess, implement)
- QMS governance, documentation & audit readiness
- Technical assessment of laboratories
- Evidence validation & data integrity

Learning, curriculum & pedagogy

- Instructional design & blended learning
- Apprenticeship & RQF curriculum (L3–L6)
- Video scripting, storyboarding & media selection
- Competence frameworks & assessment strategy
- Metacognitive & learning-science models

Education

MSc Biomedical Sciences (Immunohaematology) — Merit
Cardiff Metropolitan University · 2016

BSc (Hons) Biomedical Sciences — First Class
Cardiff Metropolitan University · 2007

BSc Biology — 2:1
University of Portsmouth · 2003

Registration

- HCPC Registered Biomedical Scientist
- Chartered Scientist (CSci)
- IBMS Full Member

Professional profile

Biomedical scientist and clinical education specialist with a strong diagnostic background across haematology, transfusion, coagulation and haemoglobinopathy screening. Over a decade in ISO 15189 and CPA-accredited NHS laboratories, progressing into technical assessment, quality systems and specialist training. Now founder of Forged Frameworks, translating scientific and regulatory rigour into accessible, high-quality learning.

70+	650+	2,000+	40%
ISO 15189 & 17025 lab assessments	Training days moved online, Feb 2020 to Mar 2021	Hours logged in the first year of Forged Frameworks, 40+ modules	Content production time cut via AI automation

Signature projects

ISLH — **Point-of-care analyser evaluation** — Comparative study of the Yumizen H500 FBC analyser for paediatric point-of-care testing, presented at ISLH.

£42k/yr — **All-Wales screening algorithm** — Redesigned sickle cell & thalassaemia antenatal screening — ~6 hours saved per patient contact.

ISO 15189 — **CPA → ISO 15189 transition** — Led change control, method validation, measurement uncertainty, budgets and quality documentation to achieve laboratory accreditation.

Professional experience

Biomedical, Diagnostic & Managerial Roles
Feb 2004 – Jul 2018

NHS Wales — Cardiff & Vale UHB

14 years in NHS pathology — from Medical Laboratory Assistant to acting senior and manager roles across haematology, transfusion and haemoglobinopathy.

Technical Assessor & Quality Specialist
Nov 2017 – 2024

UKAS

70+ ISO 15189/17025 lab assessments across healthcare, PT, forensics & calibration; UK and international.

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Further specialisms

Leadership & collaboration

- Project leadership & cross-functional coordination
- Stakeholder engagement — employers, SMEs, regulators
- Design-workshop facilitation with SMEs & curriculum teams
- Coaching, mentoring & capability development
- Commercial training delivery & consultancy

AI & digital learning

- Agentic AI authoring & content pipelines
- Bespoke HTML/CSS/JS blocks & xAPI wrappers
- Rise 360, Storyline, SCORM, xAPI, Moodle
- Accessibility (WCAG 2.2, EN 301 549)

Certifications

- TAP Diploma in Learning Design (2021)
- IBMS Certificate of Expert Practice in Training (2017)
- IBMS Specialist Diploma in Haematology & Transfusion (2007)
- UKAS Technical & Lead Assessor — ISO 15189, 22870, 17043

Platforms & tools

Laboratory & diagnostics
Automated haematology · coagulation · transfusion · haemoglobinopathy screening · EQA / IQA

Quality systems
Q Pulse · iPassport · Telepath / Dedalus · Ideagen

Authoring & media
Articulate 360 (Rise, Storyline) · SCORM · xAPI · Visme · Canva · Vyond · Descript · Camtasia

Code & AI
Python · HTML/CSS/JS · Claude Code · OpenAI Codex · OpenAI & Gemini APIs · Google Apps Script · NotebookLM

Professional experience (continued)

Assessment Manager, Medical Laboratories
July 2018 – Feb 2020

UKAS (Healthcare)

Planned and led ISO 15189 assessments for clinical and diagnostic laboratories.

Senior Training Manager
Feb 2022 – Sept 2024

UKAS Training Academy

Led trainers, budgets and the digital transformation of ISO 15189, 17025 & 17043 training.

Founder & Creator
Aug 2025 – Present

Forged Frameworks

AI-driven learning-design consultancy: 1,500+ hours of Level 3 to Level 6 modules and framework packages for Tiro Training and 150 hours of interactive medical education since September 2025, with agentic pipelines, bespoke HTML blocks and accessibility consultancy.

Learning Content Manager
Sept 2024 – Aug 2025

Tiro Training Ltd

240+ hours of L3–L6 blended learning across six technical sciences; Moodle redesign and GenAI-assisted authoring.

Training Manager
Feb 2020 – Feb 2022

UKAS Training Academy

Moved 650+ training days of the UKAS Academy portfolio to remote delivery, Feb 2020 to Mar 2021; built healthcare training content.

Founder & Blacksmith
2020 – Present

Wig's Wonky Forge

My own creative-metalwork and blacksmithing brand — hand-forged ironwork and a growing craft side-business.

Continuing professional development

GenAI in Learning Design (Coursera / Google) · Accessibility & Universal Design for Learning · ISO 15189:2022 Implementation & Internal Audit · Prompt Engineering for L&D · AI Literacy for Educators.

Values & soft skills

Collaborative · Analytical · Creative problem-solver · Reflective practitioner · Inclusive communicator · Continuous learner · Integrity-driven

Interests

Sci-fi, science and literature. Creative metalwork and blacksmithing — building my own forge and metalwork brand. Walking, running and wellbeing. Mental-health advocacy, and two very opinionated cats.

